IHTC Metamodel

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METAMODEL SUMMARY

Overview

This metamodel captures the current IHTC work in MDEO for two connected optimisation layers: patient scheduling and nurse assignment. The scheduling side decides which patients are admitted, on what day, in which room, and in which operating theatre. The nurse-assignment side then represents room-shift demand and assigns nurses to those shifts. The model also includes day-based availability objects so that transformations can update and validate state over time.

Enumerations

- **Gender.**
  Purpose: stores patient sex as `M` or `F`.
- **AgeGroup.**
  Purpose: stores patient age category as `BABY`, `CHILD`, `YOUNG`, `ADULT`, `ELDERLY`, or `EMPTY`.
  Nuance: `EMPTY` is also used in `RoomAvailability` as a sentinel value for a room-day that currently has no occupants.
- **OptimisationPhase.**
  Purpose: separates the one-way search into `PATIENTS` and `NURSES` phases.

Root Container

- **HospitalInstance.**
  Purpose: root object that collects the full optimisation state for one hospital instance.
  - **Attribute:** `decisionHorizon : int`
  - **Link:** `patients[0..*] -> Patient`
  - **Link:** `operatingtheatres[0..*] -> OperatingTheatre`
  - **Link:** `surgeons[0..*] -> Surgeon`
  - **Link:** `rooms[0..*] -> Room`
  - **Link:** `nurses[0..*] -> Nurse`
  - **Link:** `surgeonAvailabilities[0..*] -> SurgeonAvailability`
  - **Link:** `operatingTheatreAvailabilities[0..*] -> OperatingTheatreAvailability`
  - **Link:** `roomAvailabilities[0..*] -> RoomAvailability`
  - **Link:** `hospitalisationShifts[0..*] -> HospitalisationShift`
  - **Link:** `nurseWorkingShifts[0..*] -> NurseWorkingShift`
  - **Link:** `roomShiftAssignments[0..*] -> RoomShiftAssignment`

- **Link:** `deletedAdmissionsTrackers[0..*] -> DeletedAdmissionsTracker`
- **Link:** `optimisationState[1..1] -> OptimisationState`
- **Link:** `admissions[0..*] -> Admission`

Nuance: the `decisionHorizon` attribute defines the number of days in scope for the planning problem.

Patient Scheduling Core

- **Patient.**
  Purpose: represents a patient that may or may not be admitted.
  - **Attribute:** `id : int`
  - **Attribute:** `isMandatory : boolean`
  - **Attribute:** `isScheduled : boolean`
  - **Attribute:** `dueDate : int`
  - **Attribute:** `releaseDate : int`
  - **Attribute:** `ageGroup : AgeGroup`
  - **Attribute:** `surgeryDuration : int`
  - **Attribute:** `gender : Gender`
  - **Attribute:** `stayLength : int`
  - **Link:** `assignedSurgeonId[1..1] -> Surgeon`
  - **Link:** `incompatibleRooms[0..*] -> Room`
  - **Link:** `dayDemand[1..*] -> PatientDayDemand`

  Nuance: `isScheduled` is currently used as a practical modelling flag to simplify matching and transformation logic.
- **Admission.**
  Purpose: records the actual scheduling decision for a patient.
  - **Attribute:** `admissionDay : int`
  - **Link:** `patientId[1..1] -> Patient`
  - **Link:** `roomId[1..1] -> Room`
  - **Link:** `operationTheatreId[1..1] -> OperatingTheatre`

  Nuance: the admission stores only the start day, so full-stay effects are handled through day-based support objects such as `RoomAvailability`.
- **Room.**
  Purpose: represents a recovery room used for post-operative stays.
  - **Attribute:** `id : int`
  - **Attribute:** `maxCapacity : int`
- **OperatingTheatre.**
  Purpose: represents a theatre resource for surgery.
  - **Attribute:** `id : int`
- **Surgeon.**
  Purpose: represents a surgeon resource.
  - **Attribute:** `id : int`

  Nuance: surgeons are assigned to patients before optimisation rather than chosen by the transformations.

Day-Based Resource State

- **SurgeonAvailability.**
  Purpose: stores how much operating time a surgeon has on a given day.
  - **Attribute:** `day : int`
  - **Attribute:** `maxOperatingTime : int`
  - **Link:** `surgeonId[1..1] -> Surgeon`
- **OperatingTheatreAvailability.**
  Purpose: stores available theatre capacity for one theatre on one day.
  - **Attribute:** `day : int`
  - **Attribute:** `maxCapacity : int`
  - **Link:** `operatingTheatreId[1..1] -> OperatingTheatre`
- **RoomAvailability.**
  Purpose: stores room occupancy state for one room on one day.
  - **Attribute:** `day : int`
  - **Attribute:** `occupiedBeds : int`
  - **Attribute:** `ageGroup : AgeGroup`
  - **Attribute:** `roomNumber : int`
  - **Link:** `roomId[1..1] -> Room`

  Nuance: this class is central to enforcing room capacity and no-age-group-mixing constraints across a patient's full stay. The room number is duplicated here as a convenience field for transformation logic.

Nurse Assignment Layer

- **HospitalisationShift.**
  Purpose: represents a structural room-day-shift slot to which a nurse may be assigned.
  - **Attribute:** `day : int`
  - **Attribute:** `shift : int`
  - **Link:** `room[1..1] -> Room`

  Nuance: workload and required skill are derived from current `Admission` and `PatientDayDemand` records rather than stored on this class.
- **Nurse.**
  Purpose: represents a nurse resource.
  - **Attribute:** `id : int`
  - **Attribute:** `skillLevel : int`
- **NurseWorkingShift.**
  Purpose: stores when a nurse is available to work and how much workload they can cover.
  - **Attribute:** `day : int`
  - **Attribute:** `shift : int`
  - **Attribute:** `maxLoad : int`
  - **Link:** `nurse[1..1] -> Nurse`
- **RoomShiftAssignment.**
  Purpose: records the decision to assign a nurse to a particular hospitalisation shift.
  - **Link:** `nurse[1..1] -> Nurse`

- **Link:** `hospitalisationShift[1..1] -> HospitalisationShift`

Nuance: this is the final decision object for the nurse-assignment optimisation stage.

Demand and Tracking Support

- **PatientDayDemand.**
  Purpose: describes the nursing demand a patient contributes at a relative day and shift of their stay.
  - **Attribute:** `relativeDay : int`
  - **Attribute:** `shift : int`
  - **Attribute:** `workloadProduced : int`
  - **Attribute:** `skillLevelRequired : int`
  - **Link:** `patient[1..1] -> Patient`

  Nuance: together with an admission, this maps patient demand to a room-day-shift slot.
- **DeletedAdmissionsTracker.**
  Purpose: counts how many admission removals have been performed during search.
  - **Attribute:** `count : int`

  Nuance: it records deletion operations during search; the active removed-patient objective separately counts mandatory patients without admissions.
- **OptimisationState.**
  Purpose: stores the current `OptimisationPhase` for the one-way patient-then-nurse search.
  - **Attribute:** `phase : OptimisationPhase`

  Nuance: patient transformations apply in `PATIENTS`; nurse transformations apply in `NURSES` after mandatory patients are scheduled.
- **Occupant.**
  Purpose: represents a patient-room occupancy relation with a stay length.
  - **Attribute:** `stayLength : int`
  - **Link:** `patientId[1..1] -> Patient`
  - **Link:** `assignedRoomId[1..1] -> Room`

  Nuance: in the current model, most practical occupancy reasoning is handled through `Admission` and `RoomAvailability`, so this class is more auxiliary than central.

Modelling Notes

- **Decision versus derived state.** An `Admission` is the patient-scheduling decision. Room occupancy is not inferred only when needed: pre-created `RoomAvailability` records store the mutable state for each room and day of the planning horizon.
- **Full-stay occupancy.** A patient occupies their assigned room from `admissionDay` through `admissionDay + stayLength - 1`. Patient transformations validate and update every corresponding room-day record, enforcing capacity and the no-age-group-mixing rule.
- **Two optimisation phases.** `OptimisationState` begins in `PATIENTS`, where only patient transformations are enabled. Once every mandatory patient is admitted, the state changes one way to `NURSES`, which freezes patient moves and enables nurse-assignment moves.
- **Derived nursing demand.** `HospitalisationShift` is a pre-created structural room-day-shift slot. Its workload and required skill are derived from the current `Admission` objects and matching `PatientDayDemand` records, so moving an admission automatically changes the nursing demand considered by objectives and nurse transformations.
- **MDEO support fields.** `isScheduled`, `roomNumber`, and `AgeGroup.EMPTY` support matching, full-stay updates, and age-group compatibility. `Occupant` is auxiliary; active occupancy uses `Admission` and `RoomAvailability`.